

CSE490 (D)

Introduction to Football Data & Analytics

Lecture-13

2022 FIFA World Cup Final

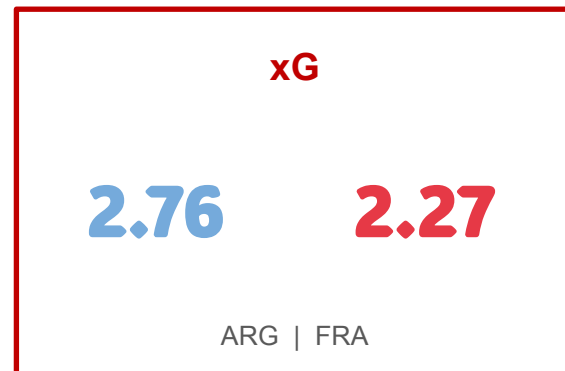
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Why This Match?

The 2022 World Cup Final is an ideal case study:

- Goals from open play, set pieces, and penalties, every shot type represented.
- Two distinct tactical approaches : Argentina's possession vs France's transitions.
- Three lead changes plus 30 minutes of extra time , multiple game states to analyse.
- Two of the world's best players , Messi vs Mbappé comparison.
- Public StatsBomb open data, every analysis here is reproducible.

The Match in Numbers



Goal scorers

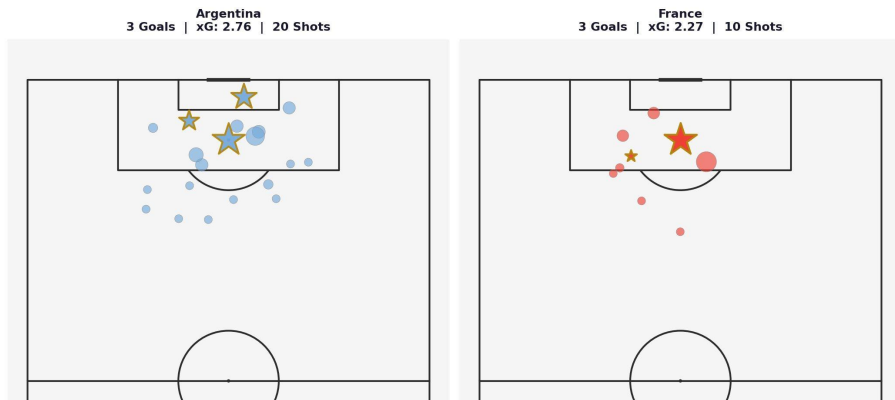
Argentina: Messi 22' (pen), Di María 35', Messi 107'

France: Mbappé 80' (pen), Mbappé 81', Mbappé 117'

3-3 after extra time. Argentina won the penalty shootout 4-2.

1. Shot Map

Shot Map — 2022 FIFA World Cup Final
(Regulation + Extra Time)

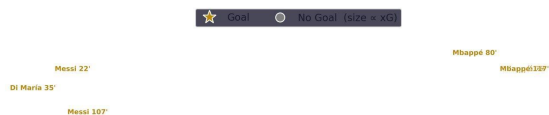


What it shows

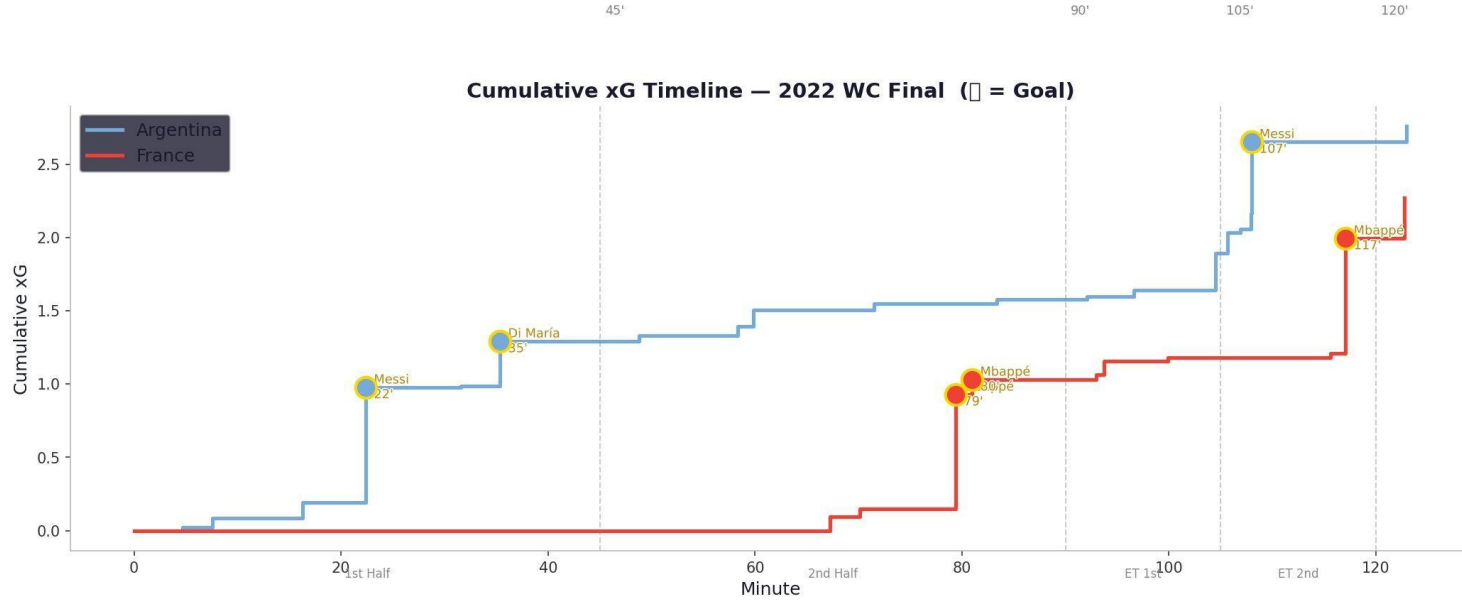
- Each circle = one shot, sized by xG.
- Stars = goals (also sized by xG).
- Position on the pitch = shot location.

Reading the chart

Argentina took 20 shots from a wide spread of locations. France's 10 shots are clustered tight in the box, fewer chances, but each one valuable. Both finished close to their xG, which is why the score ended 3-3.



2. Cumulative xG Timeline



What it shows

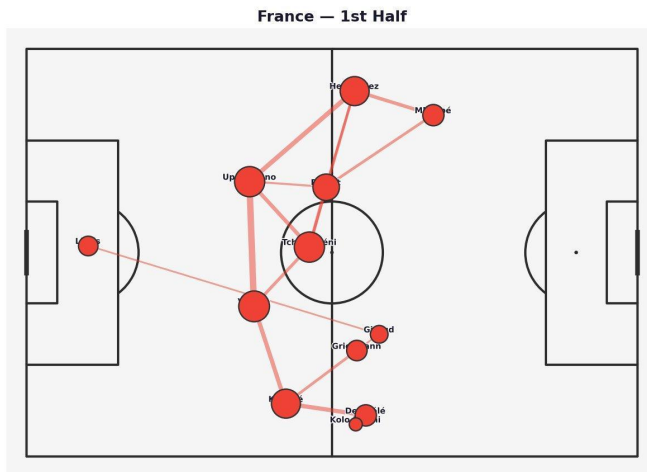
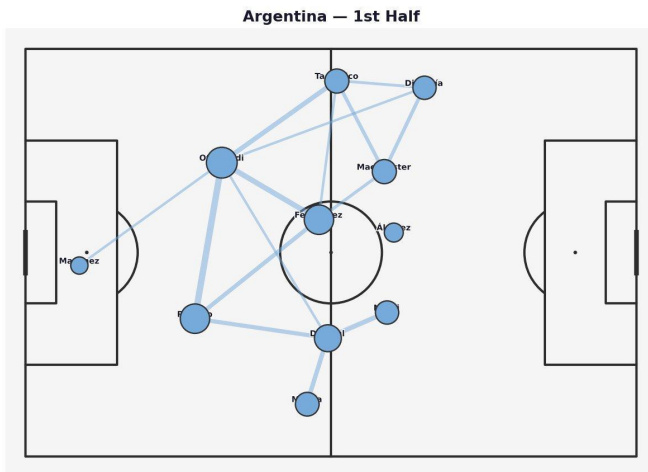
Each shot adds its xG to the running total. Vertical jumps = big chances. Yellow circles = goals. The curve shape shows when a team was creating, when it was dormant, and when momentum shifted.

2. Cumulative xG Timeline — Insights

- Argentina dominated chance-creation for 80 minutes.
- France attempted zero shots in the first half , a record for a World Cup Final.
- The 79'-81' window is the clearest tactical inflection of the match: France's flat curve goes vertical with two penalties and a goal in 90 seconds.
- Extra time produced one big chance per side , Messi 107', Mbappé 117'.
- Final xG: 2.76 vs 2.27. Argentina was the better team on chance quality.

3. Passing Network — 1st Half

Passing Network — 2022 WC Final (1st Half, Completed Passes)



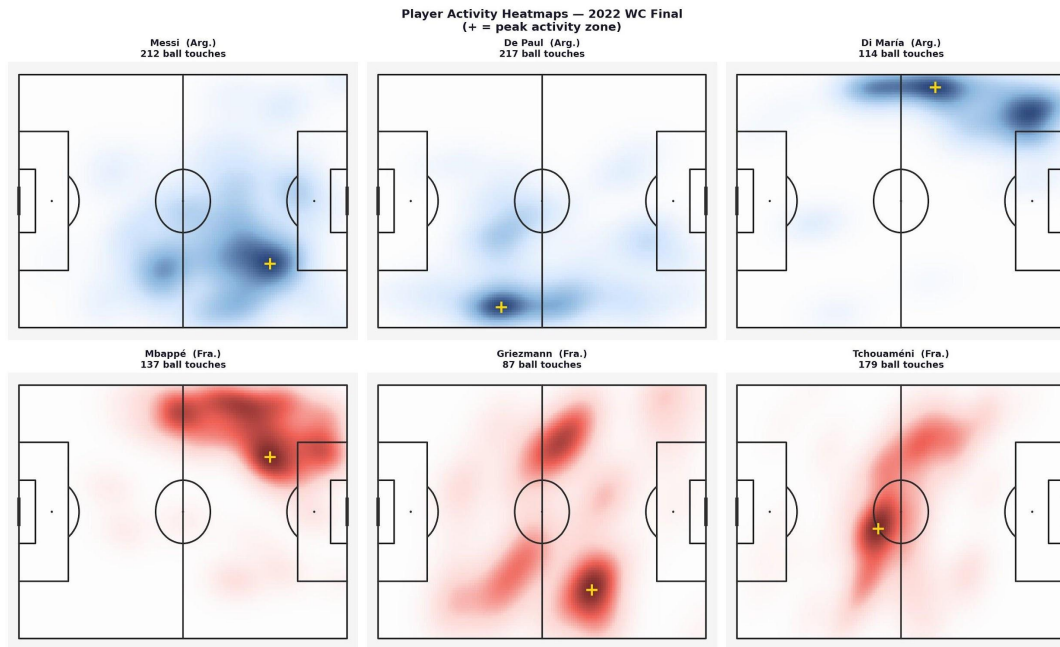
How to read it

- Node position = avg location of player
- Node size = total touches
- Edge thickness = pass volume between two players

What it tells us

Argentina's network is compact, central, and well-connected. France's is fragmented and pushed back — Mbappé isolated, no clean route from defense to attack. This is what 'no shots in the 1st half' looks like in graph form.

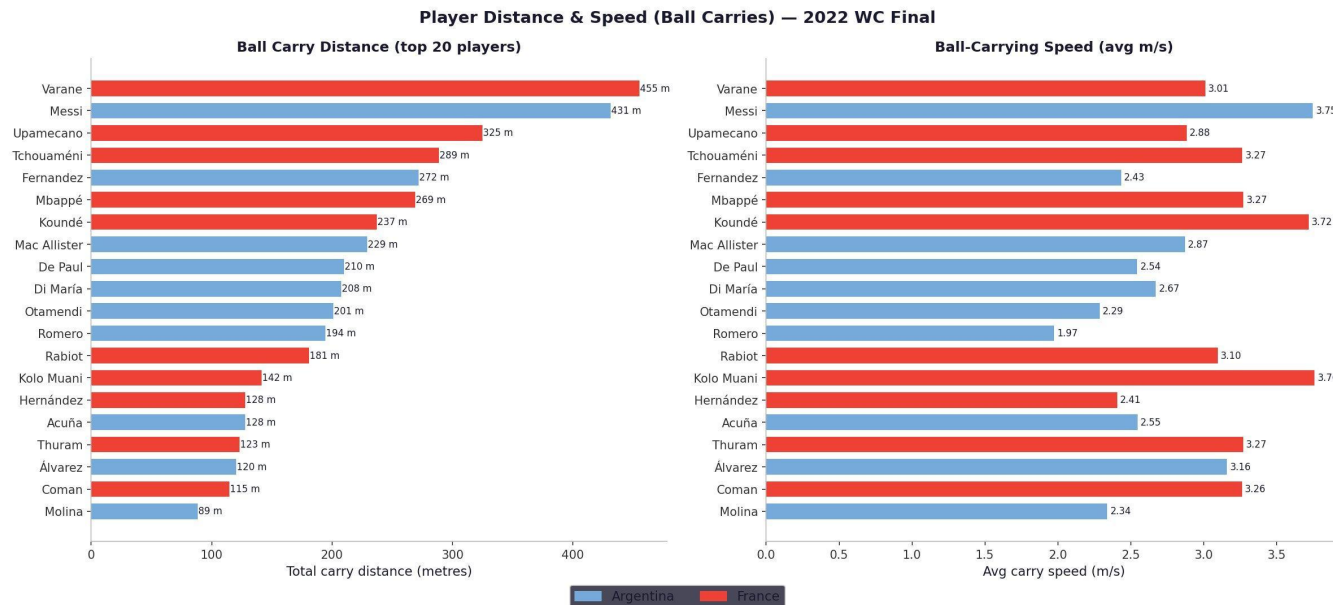
4. Player Activity Heatmaps



What it shows

Density of ball touches across the pitch for each player. Hot zones reveal tactical role at a glance. Messi roamed the right half-space (212 touches). De Paul anchored centre-left as a 6. Mbappé hugged the left flank exclusively. Tchouaméni's diagonal band marks his role as the defensive midfielder.

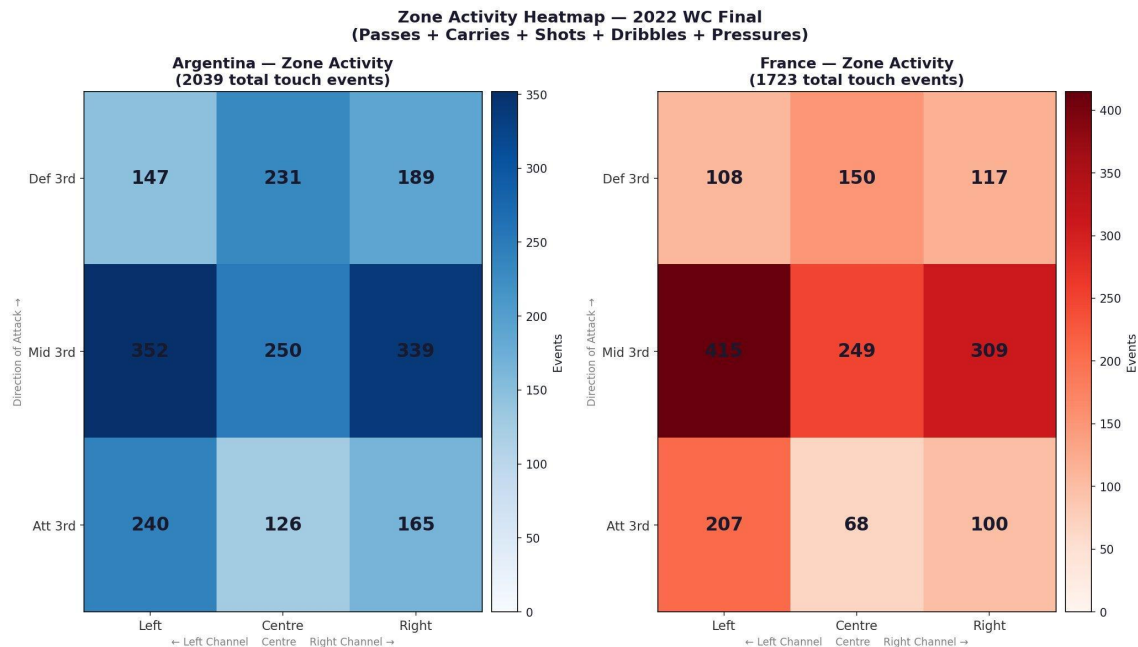
5. Ball Carries — Distance & Speed



Two charts, two questions

Distance = who progressed the ball overall (Varane 455m, Messi 431m ; heavy buildup load). Speed = who moved the ball decisively when they had it (Kolo Muani 3.76 m/s, Messi 3.75). High distance + low speed = recycler. High speed + low distance = transition runner. Read both together for the full picture.

6. Zone Activity Heatmap

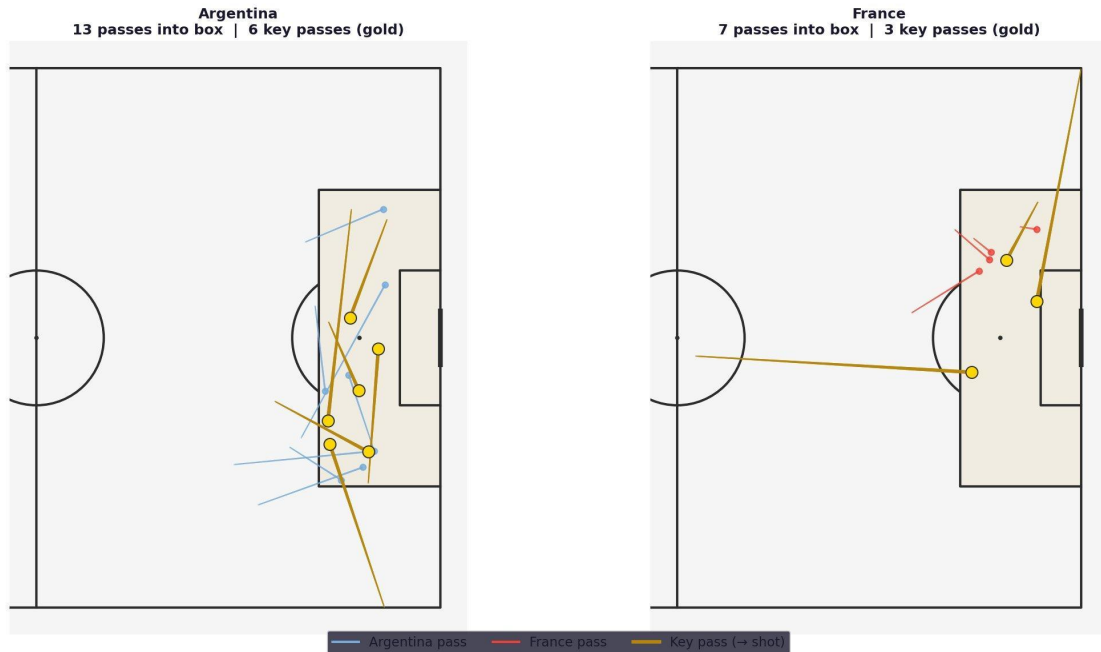


Reading the grid

Pitch divided into 9 zones (3 thirds × 3 channels). Each cell counts touch events (passes, carries, shots, dribbles, pressures). Argentina lived in the left channel (352 mid + 240 attacking). France's heaviest zone (415 in mid-3rd left) is defensive ; they were absorbing pressure, not creating from there.

7. Dangerous Passes

Dangerous Passes — Into Penalty Box | 2022 WC Final



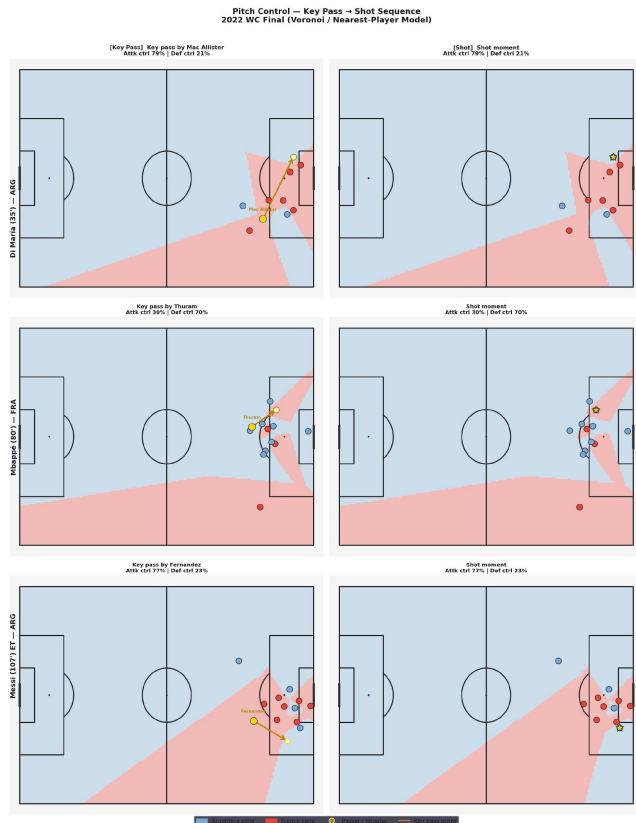
What counts as dangerous?

Any pass whose end-point lands inside the opponent's penalty box. Gold lines = key passes (the pass that immediately preceded a shot).

The numbers

- Argentina: 13 box entries, 6 key passes.
- France: 7 box entries, 3 key passes.
- Argentina almost double on both metrics.

8. Pitch Control → Key Pass → Shot



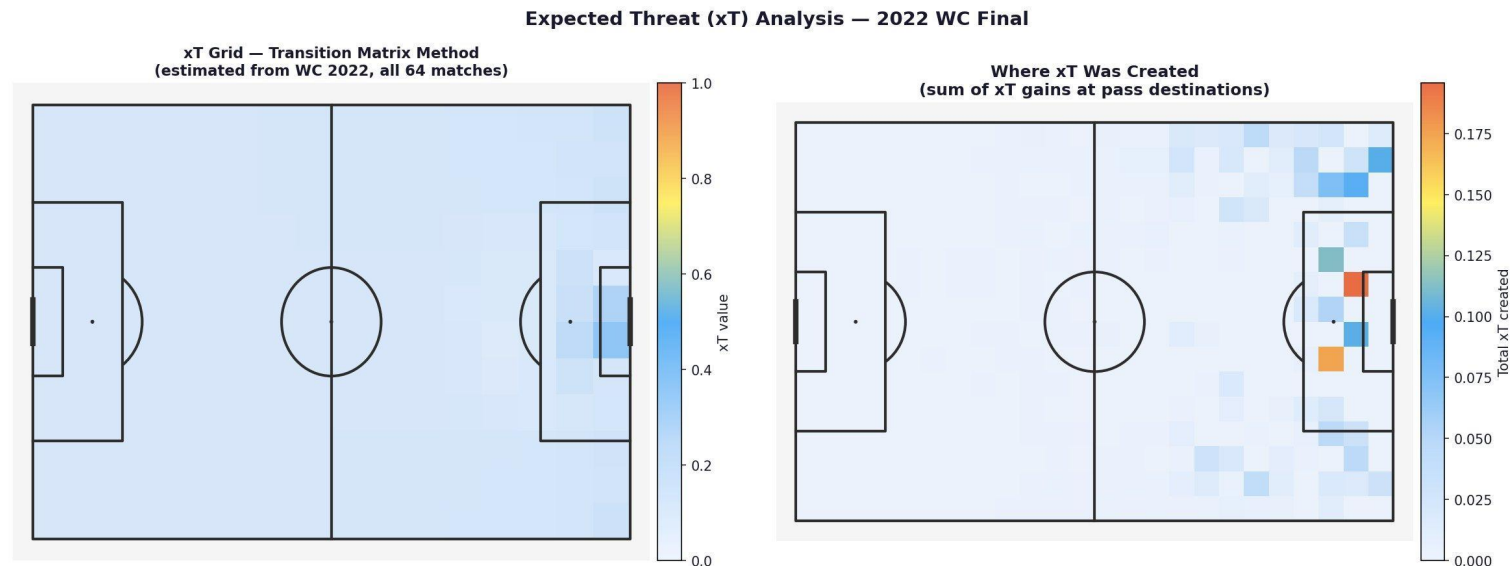
From last lecture, applied

Three key-pass moments shown twice each: at the pass and at the shot. Voronoi/nearest-player approximation of the pitch control function we built last week.

How to read

- Blue = Argentina-controlled territory.
- Red = France-controlled territory.
- Yellow arrow = the key pass.
- Star (right panel) = where the shot was taken.

9. Expected Threat (xT) Grid

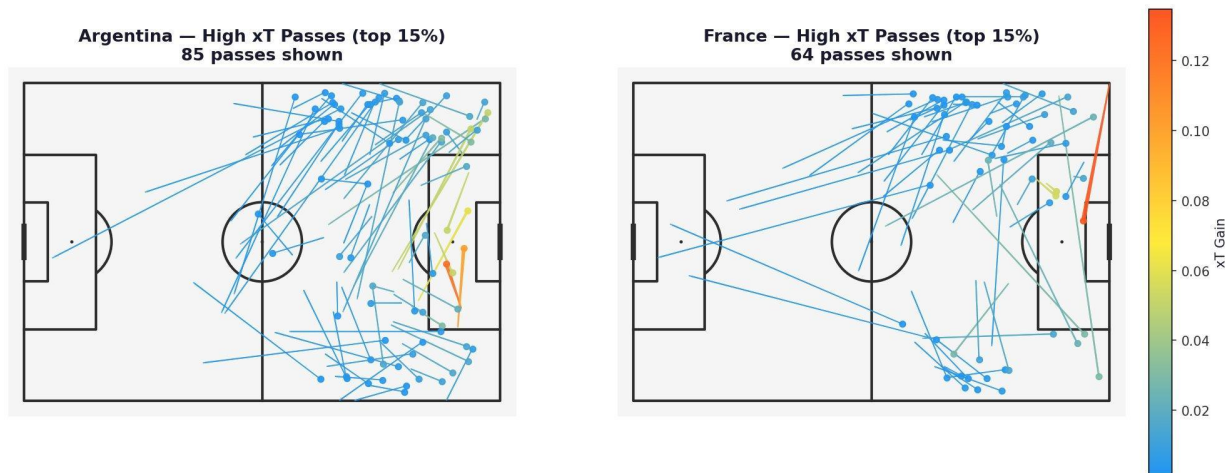


Two views of position value

- Left: the xT grid itself — for each cell, the probability that a possession reaching it ends in a goal within a few actions. Estimated from all 64 WC 2022 matches.
- Right: where xT was actually created in the final, summed over pass destinations. The hot orange cell is where Mac Allister's key pass landed.

10. High-Threat Passes — Top 15%

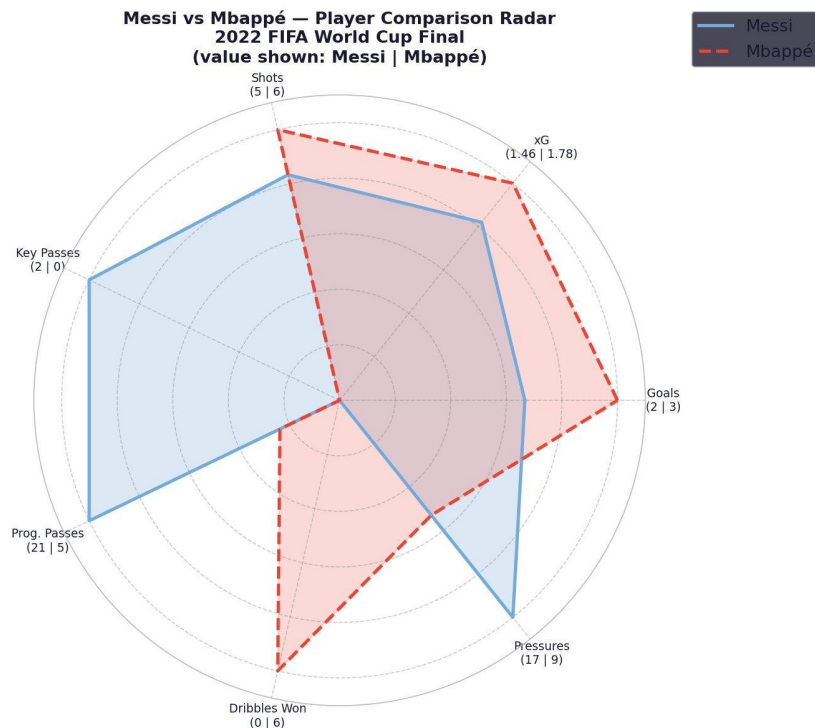
High Expected Threat Passes — 2022 WC Final



Volume vs concentration

Argentina played 85 passes in the top-15% xT bracket; France only 64. But notice France's single deep-orange line ; one extremely high-xT pass into the right channel. Argentina spread their threat across many passes; France concentrated theirs into fewer, higher-value ones. Two different attacking signature..

11. Messi vs Mbappé



Two ways to dominate a final

Mbappé: pure finisher

6 shots, 6 dribbles won, 3 goals ; but 0 key passes and only 5 progressive passes. End-of-chain striker who relied on teammates to deliver the ball.

Messi : creator + presser

21 progressive passes, 2 key passes, 17 pressures, plus 5 shots and 2 goals. Involved in the build-up, the chance creation, and the defensive press.

What 11 Charts Together Tell Us

Argentina's match

- Sustained territorial control through 80'
- Left-channel attacking pattern (Mac Allister, Acuña, Di María)
- Higher xG, shot volume, and box entries
- Messi as creator-presser, not just finisher

France's match

- Reactive, transition-based — fragmented passing network
- First-half shutout, then clinical 90-second flurry at 79'-81'
- Concentrated threat — fewer passes, higher value
- Mbappé as pure finisher, Tchouaméni / Hernández did the build-up